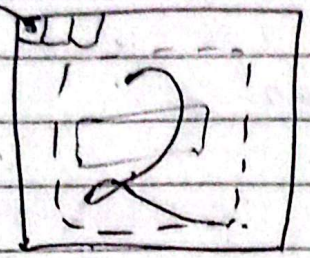
Curse of Dimensionality.

features → dimensions

There should be optimal number of feature, adding more will not be beneficial.

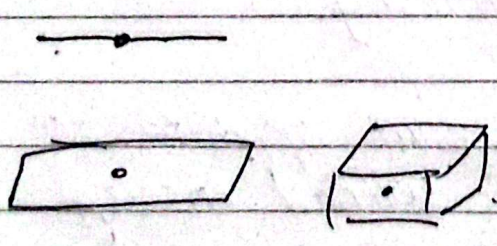
784 px. 28*28.



- working on will be bad.
- working on will be best
- working on more than req^d.

$f_1 \rightarrow f_{784} \} 784 - D$

- so take optimum number of features.



} finding anything will be difficult as increase in the dimension.

- KNN will fail

solution { Dimension Reduction }

f_1 - - - - f_{1000}.

Page No.
Date: / /

Dimension Reduction

Feature Selection

$f_1, f_{10}, f_{20}, f_{22}, f_{500}$.

types:
- Forward selection
- Backward selection
- Bidirection elemination

mostly use.

Feature Extraction

~~type~~ techniques:
1) PCA.
make another columns which are the linear combinations of other columns.
2) LDA
3) t-SNE

Feature Extraction.

1) PCA [Principle Component Analysis]
- unsupervised ml problem.
- complex technique.
- It is feature extraction technique to reduce curse of dimensionality.
- reduces higher dimension of data to lower dimension of data without lossing the essence of the data.

Benefits:
- faster execution of algorithm.
- visualization.

Page No.
Date: / /

Geometric Intuition.

No. of rooms	No. of near grocery shop	price (2)
3	2	60
4	0	130

Feature selection : selecting important calures only.

there are two features.

i) No. of rooms

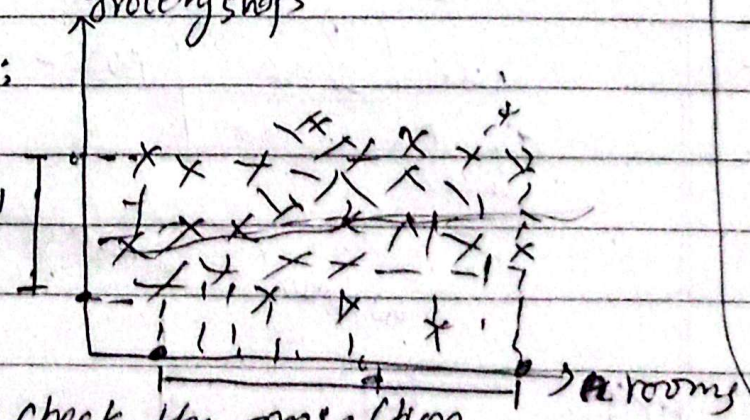
ii) No. of near grocery shop.

- Let's say we have to remove one from here.
- If you have domain knowledge, it is good
  i) No. of rooms.

If you don't have domain knowledge

Then: . Groceryshops

- Step1:



$d > d'$

- var(). Check variance
- choose of higher variance.

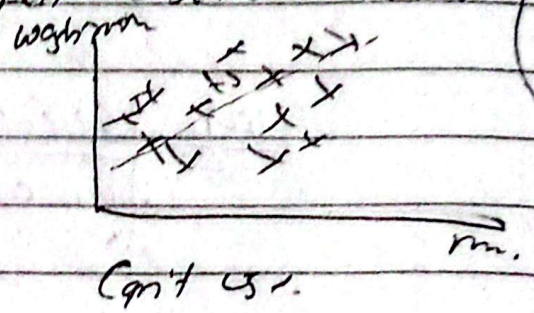
- Step 2: check the projection.
  remove the lower projection one,
  & select data sparate.

Dataset 2:

No. of room	No. of bathroom	flat price

- Both imp and can't use feature selection here.

$d' \simeq d$



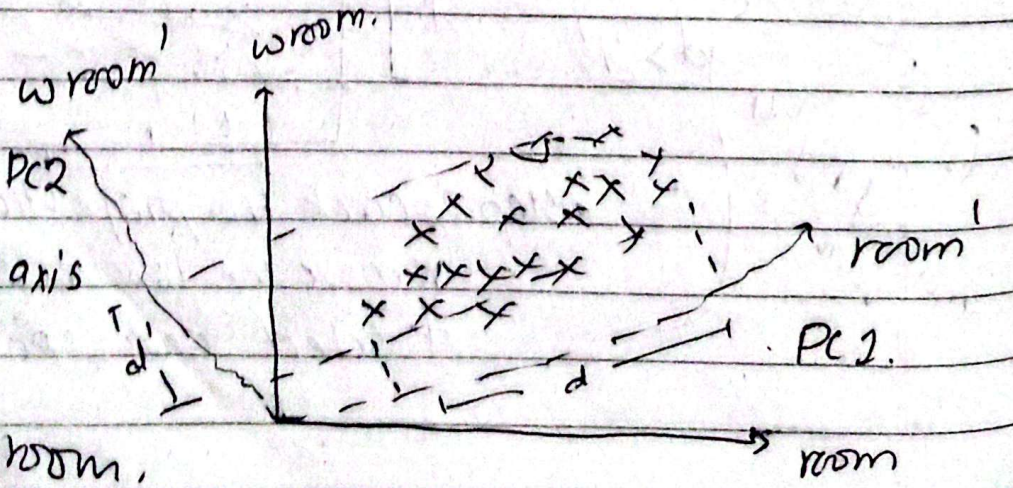
Can't use.

How to solve this by feature extraction.

room	w. room	price

use different variable.

size	price



PCA now shift the axis

$d > d'$

& remove wroom.

Principle Component 1.

PC1 no. of PC ≤ no. of original features in data.

Variance [measures the spread of data]

- Small variance → values are close to mean
- Large variance → values are spread out.

Variance ∝ spread.

1) Population Variance [σ^2]

↳ if there are 100 students' data take all.

$$\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$$

μ = true mean

Division = N

2) Sample Variance (s^2)

↳ take few sample data: 10 students' data

$$s^2 = \frac{\sum_{i=1}^{N} (x_i - \bar{x})^2}{N-1}$$

$\bar{x}$ = sample mean.

Division = N-1.

Normally individual data (-5, 0, 5)

$$\sigma^2 = \frac{(-5)^2 + 0^2 + 5^2}{3} = \frac{50}{3}$$

Page No.
Date: / /

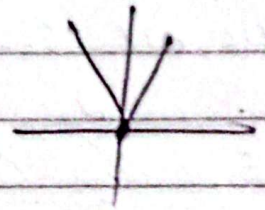
(2) Why not to use Mean Absolute Deviation (MAD)? Why variance.

→ $MAD = \frac{\sum |x_i - \bar{x}_i|}{N}$

- It tells how far, on avg, the data is from the mean, without caring about direction.

Can't differentiate betn $\underbrace{-2 \ \& \ 2}_{\text{misleading}}$ } absolute.

Graph of mod values



Can't be differentiable in 0.
~~which~~ ~~we~~ we can't use optimization later in PCA.

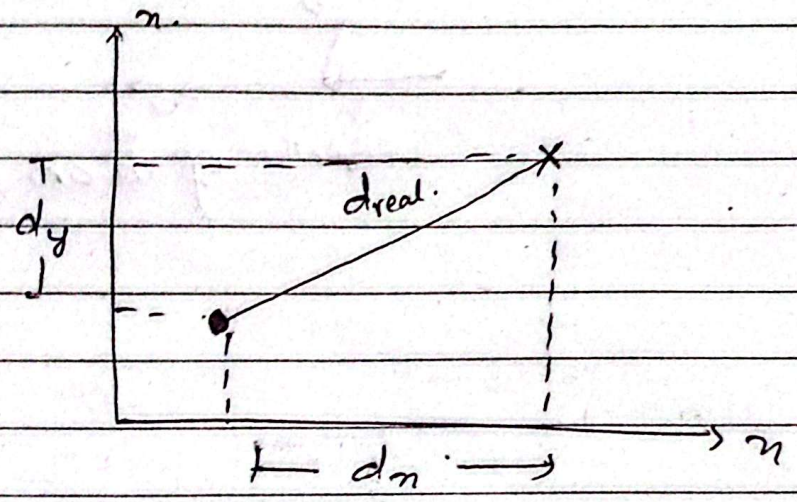
→ Variance sq. function is differentiable & optimization can be used.

Page No.
Date: / /

Why variance is important?

→ when will do feature extraction (reduction)

if we will project in wrong axis, then distance based algorithm will be of high error. [so, we can't use].



$d_{real} \approx d_x$

but $d_{real} >> d_y$

hence, distance based algorithm will be affected much and accuracy will be very much affected.